

Resource and parsing layer

PDF full text

Metadata & license

Figures, tables & equations

Citations & linked data

Parser version, page order & object positions

User task requirements

Shared scientific-information substrate

Unified scientific-document representation

Clean text & section structure

Stable object IDs & provenance

Paper / page / bounding box / section / paragraph

Caption-text links & object context

Entities, methods, data, claims & qualifiers

Design principle: preserve shared scientific information once; derive task-specific views later.

Task-conditioned application views and decisions

Research question

RQ1 Representation

How should evidence be organized for scientific RAG?

Operationalization

Fixed chunks
Atomic units
Evidence packages

Decision / metric

CES@5
BudgetCES

RQ2 Source-paper prioritization

Which source papers should be processed first?

Task-specific yield prediction
NB2 + 5-fold OOF

Paper-budget / token-budget capture

RQ3 Training-set sizing

How much machine-verified supervision should be used for training?

Nested subsets
Practical equivalence
Model replication

Practical-equivalence size n^*